

TRICKS

OTICA GEOMETRICA:

$$M_{sq.} = \begin{pmatrix} 1 & 0 \\ \frac{1}{R}(\frac{n}{n'}-1) & \frac{n}{n'} \end{pmatrix}$$

$$M_{pr.} = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

$$M_{tot} = M_n M_{n-1} \dots M_2 M_1$$

$R > 0$ lente convessa (, $R < 0$ lente concava)

$$M_{lente} = \begin{pmatrix} 1 & 0 \\ \frac{n_2-n_1}{n_2}(\frac{1}{R_2}-\frac{1}{R_1}) & 1 \end{pmatrix}$$

$$-\frac{1}{f} = \frac{n_2-n_1}{n_2} \left(\frac{1}{R_2} - \frac{1}{R_1} \right)$$

$$\det M_{tot} = \frac{n_0}{n_2}$$

$$IN \quad f_1 = \frac{n_0}{n_2} \frac{1}{c} \quad (\overrightarrow{PP_1-F_1})$$

$$OUT \quad f_2 = -\frac{1}{c} \quad (\overrightarrow{PP_2-F_2})$$

$$p = \frac{R}{c} \quad (\overrightarrow{IN-F_1})$$

$$q = -\frac{A}{c} \quad (\overrightarrow{OUT-F_2})$$

$$r = \frac{D-M_0/M_2}{c} \quad (\overrightarrow{IN-PP_1})$$

$$s = \frac{1-A}{c} \quad (\overrightarrow{OUT-PP_2})$$

$$v = \frac{D-1}{c} \quad (\overrightarrow{IN-N_1})$$

$$w = \frac{m/m_2-A}{c} \quad (\overrightarrow{OUT-N_2})$$

- Esercizio:
- matrice di trasferimento
 - check
 - punti cardinali e piani principali
 - immagine $[|K|=0] \rightarrow \begin{cases} x > 0 & \text{imm. reale} \\ x < 0 & \text{imm. virtuale} \end{cases}$

OTICA DIFFRAZIONE

1) Hf. approssimazione Fraunhofer $z \gg \frac{k}{2}(x_0^2+y_0^2)/k$

$$U(x_0, y_0) = \frac{e^{j k z}}{j \lambda z} e^{j \frac{k}{2z}(x_0^2+y_0^2)} \iint_{-\infty}^{\infty} U(x, y) e^{j \frac{k}{2z}(x^2+y^2)} e^{-j \frac{k}{z}(x_0 x + y_0 y)} dx dy$$

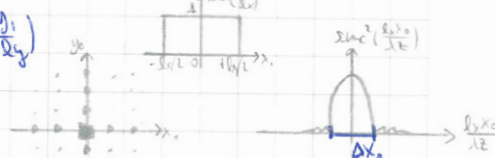
$$= \frac{e^{j k z}}{j \lambda z} e^{j \frac{k}{2z}(x_0^2+y_0^2)} \iint_{-\infty}^{\infty} U(x, y) e^{-j 2 \pi (f_x x + f_y y)} dx dy$$

$$\propto \mathcal{F}\{U(x, y)\} \quad \text{con } f_x = \frac{x_0}{\lambda z} \quad \text{e } f_y = \frac{y_0}{\lambda z}$$

- aperture rettangolare $t(x, y) = \text{rect}(\frac{x}{l_x}) \text{rect}(\frac{y}{l_y})$

$$\mathcal{F}\{t\} = l_x l_y \text{sinc}(l_x f_x) \text{sinc}(l_y f_y)$$

$$\Delta x_0 = 2 \bar{x}_0 = \frac{2 \lambda z}{l_x} \rightarrow \Delta f_x = \frac{\Delta x_0}{\lambda z} = \frac{2}{l_x}$$



- aperture circolare $t(r) = \text{circ}(\frac{r}{R})$

$r^2 = x^2 + y^2$, R = diametro apertura

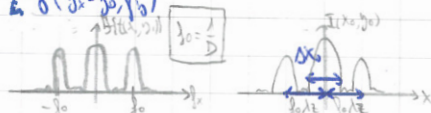
$$\mathcal{F}\{t\} = \left(\frac{R}{\lambda z}\right)^2 \left[2 \frac{J_1(\pi R f)}{\pi f} \right]$$

$$\Delta x_0 = 1,22 \frac{\lambda z}{R}$$

- aperture graduate + reticolo di diffrazione $t(x, y) = \left[\frac{1}{2} + \frac{a}{\pi} \cos(2\pi f_x x) \right] \text{rect}(\frac{x}{l_x}) \text{rect}(\frac{y}{l_y})$

$$\mathcal{F}\{t\} = \frac{1}{2} \delta(f_x) + \frac{a}{\pi} \delta(f_x + f_0) + \frac{a}{\pi} \delta(f_x - f_0)$$

$$\mathcal{F}\{t\} = l_x^2 \text{sinc}(l_x f_x) \text{sinc}(l_y f_y)$$



$$\mathcal{F}\{t\} = \mathcal{F}\{t_0\} * \mathcal{F}\{t_g\} = \frac{l_x^2}{2} \text{sinc}(l_x f_x) \left[\text{sinc}(l_y f_y) + \frac{a}{2} \text{sinc}(l_y(f_y + f_0)) + \frac{a}{2} \text{sinc}(l_y(f_y - f_0)) \right]$$

$$\Delta x_0 = \frac{2 \lambda z}{l_x} \quad f_0 \lambda z > \Delta x_0 \rightarrow f_0 > \frac{2}{l_x}$$

$$k_x = 2\pi f_x = 2\pi \frac{x_0}{\lambda z} = 2\pi \frac{f_0 l_x}{\lambda z} = 2\pi f_0$$

$$k_y = 2\pi f_y = 2\pi \frac{y_0}{\lambda z} = 0$$

$$\theta = \frac{x_0}{z} = \frac{f_0 \lambda z}{z} = f_0 \lambda \quad [\text{in rad}]$$

angolo d'incidenza rispetto a z

ToF nel fuoco

2) Campo nella regione di Fraunhofer $U(x_0, y_0) = \frac{e^{jkt}}{j\lambda z} e^{j\frac{k}{2z}(x_0^2 + y_0^2)} \iint U(x, y) e^{j\frac{k}{2z}(x+x_0)^2} e^{-j\frac{k}{2z}(y+y_0)^2} dx dy$

Le lenti associano l'immagine da distanza infinita a distanza focale, compiendo uno spostamento

$$\phi(x, y) = km\Delta(x, y) + k[\Delta_0 - \Delta(x, y)] = k\Delta_0 + k(m-1)\Delta(x, y)$$

$$\Delta(x, y) = \Delta_0 - \frac{x^2 + y^2}{2} \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad \frac{1}{g} = (m-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\rightarrow U_2 = U_1 t(x, y) = U_1 e^{j\phi(x, y)} = U_1 e^{jkm\Delta_0} e^{-j\frac{k}{2g}(x^2 + y^2)}$$

Poniamo la lente in $(x, y) = (x, y)$

$$U(x_0, y_0) = \frac{e^{jkt}}{j\lambda z} e^{j\frac{k}{2z}(x_0^2 + y_0^2)} \iint U_1 e^{-j\frac{k}{2g}(x^2 + y^2)} e^{j\frac{k}{2z}(x+x_0)^2} e^{-j\frac{k}{2z}(y+y_0)^2} dx dy$$

$$U(x_0, y_0) = \frac{1}{j\lambda g} e^{j\frac{k}{2g}(x_0^2 + y_0^2)} \iint U_1 e^{-j\frac{k}{2g}(x^2 + y^2)} dx dy = \frac{1}{j\lambda g} e^{j\frac{k}{2g}(x_0^2 + y_0^2)} F_2(f_x, f_y)$$

Troviamo $F_2(f_x, f_y)$ lavorando nelle frequenze e spostando le lenti a d_0 dall'apertura

$$F_2 = F_0 e^{j\frac{k}{2g}(x_0^2 + y_0^2)} = F_0 e^{j\frac{k}{2g} \frac{\lambda}{2} (f_x^2 + f_y^2) d_0} = F_0 e^{-j\frac{\pi}{2} \frac{\lambda}{\lambda} (f_x^2 + f_y^2) d_0} = F_0 e^{-j\frac{\pi}{2} \frac{1}{f_x^2 + f_y^2} d_0}$$

$$U(x_0, y_0) = \frac{1}{j\lambda g} e^{j\frac{k}{2g}(x_0^2 + y_0^2)} F_2 = \frac{1}{j\lambda g} e^{j\frac{k}{2g}(1 - \frac{d_0}{g})(x_0^2 + y_0^2)} F_0 = \frac{1}{j\lambda g} F_0 \left(\frac{x_0}{\lambda g}, \frac{y_0}{\lambda g} \right)$$

RISOSTO IMPULSIVA LENTE

3) Effetto Whittaker (lenti alle dimensioni finite dell'apertura del sistema ottico).
 $U_0(x_0, y_0) \rightarrow h(x_i, y_i; x_0, y_0) \rightarrow U_1(x_0, y_0)$ relazione lineare
 h , FFD del sistema, è specialmente invariante in regime parassiale
 $\Rightarrow$ integrale di sovrapposizione $\rightarrow$ integrale di convoluzione
 la perfetta riproduzione dell'immagine se $h \sim \delta(x_0 \pm \pi x_0, y_0 \pm \pi y_0)$

Effetto Whittaker lente: $P(x, y) = \begin{cases} 1 & \text{lente trasparente} \\ 0 & \text{lente opaca} \end{cases}$ e P.O.P. non viene

$$U(x_0, y_0) = \iint U_1(x, y) h(x_0 - x, y_0 - y) dx dy$$

U_0

$$U_1 = \frac{1}{j\lambda \pi} e^{jkt} \approx \frac{1}{j\lambda d_0} e^{j\frac{k}{2d_0} [(x-x_0)^2 + (y-y_0)^2]}$$

$$U_1'(x, y) = U_1(x, y) P(x, y) e^{-j\frac{k}{2g}(x^2 + y^2)} \quad (f > 0)$$

$$h(x_0, y_0; x_0, y_0) = \frac{1}{j\lambda d_0} \iint U_1'(x, y) e^{j\frac{k}{2d_0} [(x-x_0)^2 + (y-y_0)^2]} dx dy$$

$$= \frac{1}{\lambda^2 d_0^2} e^{j\frac{k}{2d_0}(x_0^2 + y_0^2)} e^{j\frac{k}{2d_0}(x_0^2 + y_0^2)} \iint P(x, y) e^{j\frac{k}{2} \left(\frac{1}{d_0} + \frac{1}{d_0} - \frac{1}{g} \right) (x^2 + y^2)} e^{-j\frac{k}{2} \left(\frac{x_0^2 + y_0^2}{d_0} + \frac{x_0^2 + y_0^2}{d_0} \right)} dx dy$$

$$= \frac{1}{\lambda^2 d_0^2} e^{j\frac{k}{2d_0}(x_0^2 + y_0^2)} e^{j\frac{k}{2d_0}(x_0^2 + y_0^2)} \iint P(x, y) e^{-j\frac{k}{2} \left(\frac{1}{d_0} + \frac{1}{d_0} - \frac{1}{g} \right) (x^2 + y^2)} dx dy$$

$$= \mathcal{F}\{P(x, y)\} \quad f_x = \frac{\pi x_0 + x_0}{\lambda d_0}, \quad f_y = \frac{\pi y_0 + y_0}{\lambda d_0}$$

$$h(f_x, f_y) = \mathcal{F}\{h(x, y)\} = \mathcal{F}\{\mathcal{F}\{P(x, y)\}\} \xrightarrow{\text{relazione di Poisson}} P(x, y)$$